

Lab # 2: Introduction to Cisco Packet Tracer

Outline

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Introduction:

Cisco Packet Tracer is a powerful network simulation and visualization tool that lets you design, configure, and test networks in a virtual environment. It helps sharpen skills for job interviews, certification exams, and IoT applications. By simulating real-world setups with unlimited devices, Packet Tracer encourages discovery, troubleshooting, and hands-on practice. It builds critical thinking, problem-solving, and decision-making skills while complementing classroom learning and Networking Academy curricula.

Setup

1. Download the student version of Cisco Packet Tracer from FileHorse:
[**https://www.filehorse.com/download-cisco-packet-tracer-32/27899/download/**](https://www.filehorse.com/download-cisco-packet-tracer-32/27899/download/)
2. Install the software using the installation wizard.
3. Launch Packet Tracer and explore the environment.

Packet Tracer Interface

Area 1: Menu bar – Common menu to open, save, print, and change preferences.

Area 2: Main toolbar – Shortcut icons like open, save, zoom, undo, redo.

Area 3: Logical/Physical workspace tabs – Toggle between logical and physical work areas.

Area 4: Workspace – Area to create topologies and view simulations.

Area 5: Common tools bar – Controls to manipulate topologies (select, move, delete, notes, etc.).

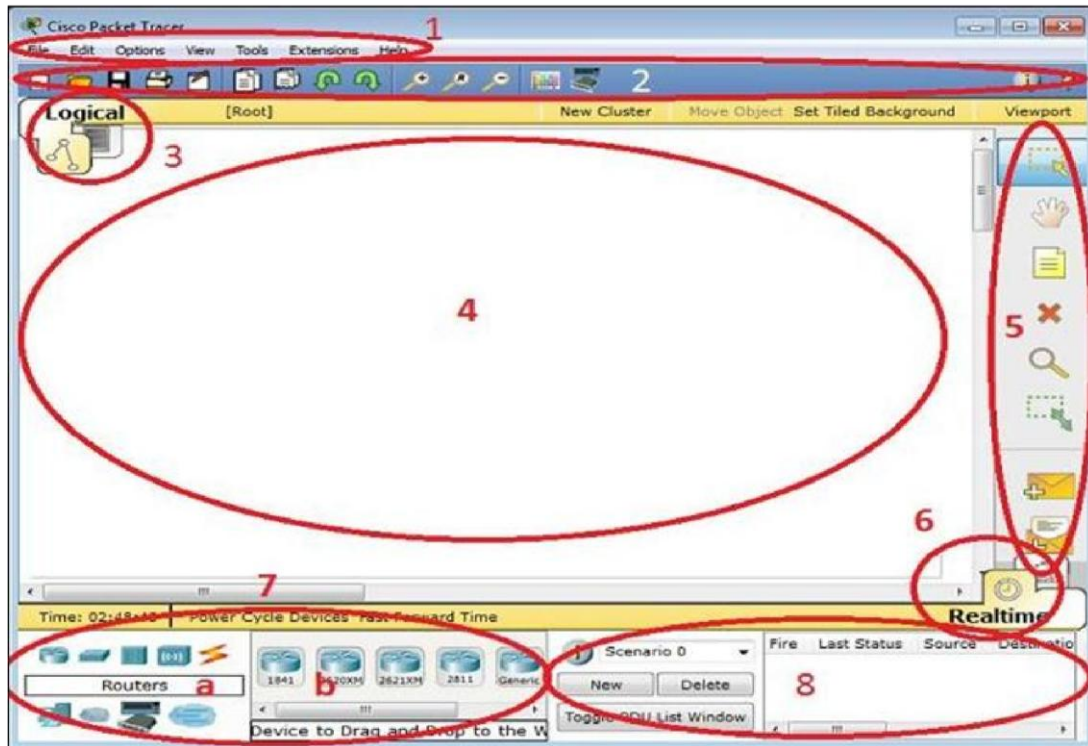
Area 6: Real-time / Simulation tabs – Toggle between real-time and simulation modes.

Area 7: Network component box – Contains all network/end devices, divided into:

- 7a: Device-type selection box
- 7b: Device-specific selection box

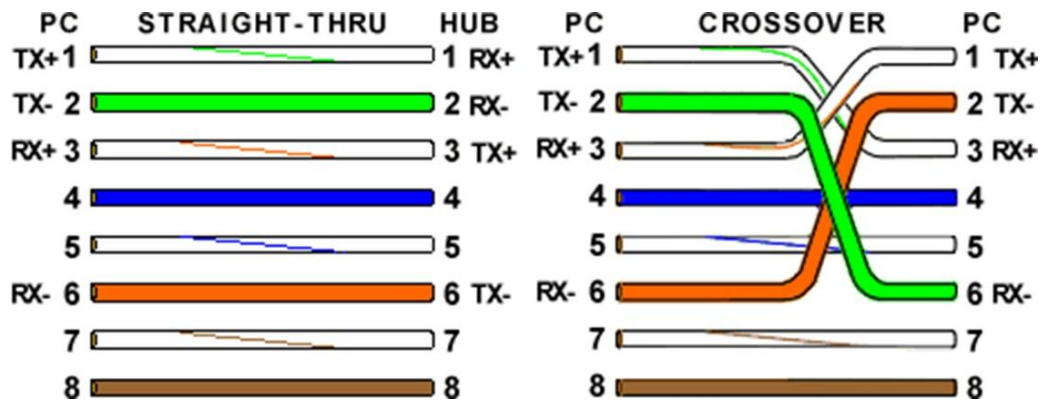
Area 8: User-created packet box – Custom packet creation and analysis.

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Cables and Connectors

- ✦ Cisco Packet Tracer provides a variety of cables and connectors to simulate real networking setups, including:
- Straight-through cables (used to connect different devices, e.g., PC to switch).
 - Crossover cables (used to connect similar devices, e.g., PC to PC, switch to switch).
 - Serial cables for router interconnections.
 - Fiber optic cables for high-speed connections.
 - Console cables for device configurations



Lab Tasks

Task 1: Explore Packet Tracer Interface

- Identify all the interface areas (Menu, Toolbar, Workspace, Device boxes, Simulation, etc.). - Take a screenshot of each area and label them.

Task 2: Connect Devices with Cables

- Place two PCs in the workspace.
- Connect them using a crossover cable.
- Replace the cable with a straight-through cable and note the difference.

Task 3: Assign IP Addresses

- Assign IP address 192.168.1.1 to PC0.
- Assign IP address 192.168.1.2 to PC1.
- Test connectivity by sending a ping command.

Task 4: Experiment with Simulation Mode

- Send a packet in real-time mode.
- Switch to simulation mode and observe the step-by-step packet flow.
- Record your observations.

Lab Questions

1. What are the main features of Cisco Packet Tracer?
2. Differentiate between crossover and straight-through cables.
3. What do you mean by T568A and T568B standards?

4. What is the function of the logical layer and physical layer in Packet Tracer?
5. In how many ways can a network be tested in Packet Tracer? Explain them.

Answers:

1. Cisco Packet Tracer is a network simulation tool. Its main features include:

- Creating network topologies
- Configuring routers and switches
- Simulating network traffic
- Testing connectivity
- Troubleshooting networks
- Supporting real-time and simulation modes

2. **Straight-through cable:** Used to connect different types of devices, such as a PC to a switch.

Crossover cable: Used to connect similar devices, such as a PC to a PC or switch to a switch.

3. T568A and T568B are wiring standards that define the color order of wires inside an Ethernet cable.

T568A: Green pair comes first.

T568B: Orange pair comes first.

T568B is commonly used in Ethernet networks.

4. **Logical layer:** Shows how devices communicate and how data moves through the network.

Physical layer: Shows the actual devices, cables, ports, and physical connections.

5. There are **two main ways**:

1. Real-Time Mode: Shows network activity as it happens. It is useful for checking connectivity.

2. Simulation Mode: Shows packets moving step by step through the network. It is useful for analyzing and troubleshooting data flow.